UNM & CNM APP CONTEST 2026

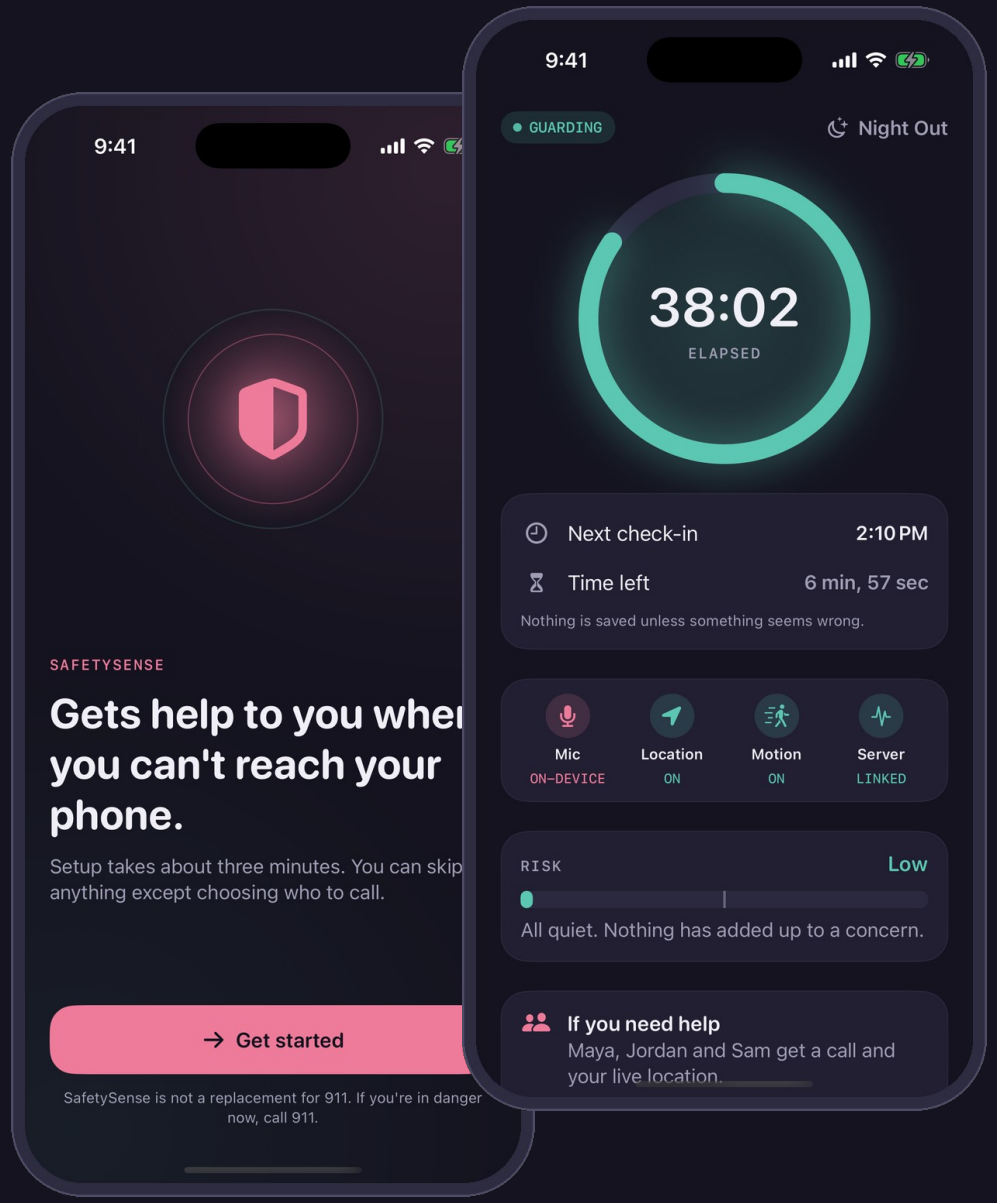
# SafetySense

## Calls for help when you can't.

A proactive personal-safety app for UNM and CNM students. It notices when something is wrong and brings in your people, then professional dispatch, even if you can't reach your phone.

**George Teifel** UNM **Mary Teifel** CNM

*SafetySense supplements 911. It never replaces it.*



WHO WE ARE

# Two siblings, two campuses, one problem

## George Teifel

University of New Mexico

Senior · Electrical Engineering

Built the iOS app and the backend

## Mary Teifel

Central New Mexico Community College

Freshman · Mechanical Engineering

Research, UX and the app's wording

THE APP IN 30 SECONDS

You start a Guard session for a night out, a walk home, a date or a ride. SafetySense listens and watches on your phone only. If you go quiet, can't answer, or your phone goes dark, it calls your people with a live map and then brings in professional dispatch.

- Native iOS 26 app: Swift 6 and SwiftUI
- On-device AI: SoundAnalysis, SpeechAnalyzer, Foundation Models
- Supabase, Twilio voice, Noonlight dispatch (sandbox)

THE PROBLEM

# Most safety apps assume you can press a button

Our users: UNM and CNM students, 18 to 25, out at night at parties, bars, first dates, rideshares and the walk back to the car.

**70%**

of women in a clinical study of assault survivors reported significant tonic immobility: involuntary paralysis.

*A panic button does nothing for someone who can't move.*

Möller et al., Acta Obstet Gynecol Scand, 2017 (n = 298)



## Drugging takes away the ability to act

Someone who has been drugged may not recognize danger or be able to unlock a phone. No phone can detect a drug, but it can notice that you stopped responding.



## The phone gets taken or thrown

A timer can't tell a dead phone from a quiet night until it runs out. SafetySense's server treats a lost heartbeat as a warning sign.




## New Mexico still has no text-to-911

It's still listed as the only US state without it. The statewide 911 upgrade began in 2025, but if you can't speak, you can't just text 911 here.

OUR ANSWER

# It watches, so you don't have to tap

1

## It only listens when you turn it on

Night Out, Walk Home, Date or Ride. Clear start, clear end.

2

## Nothing is saved unless something's wrong

Audio lives in a 30-second memory buffer that overwrites itself.

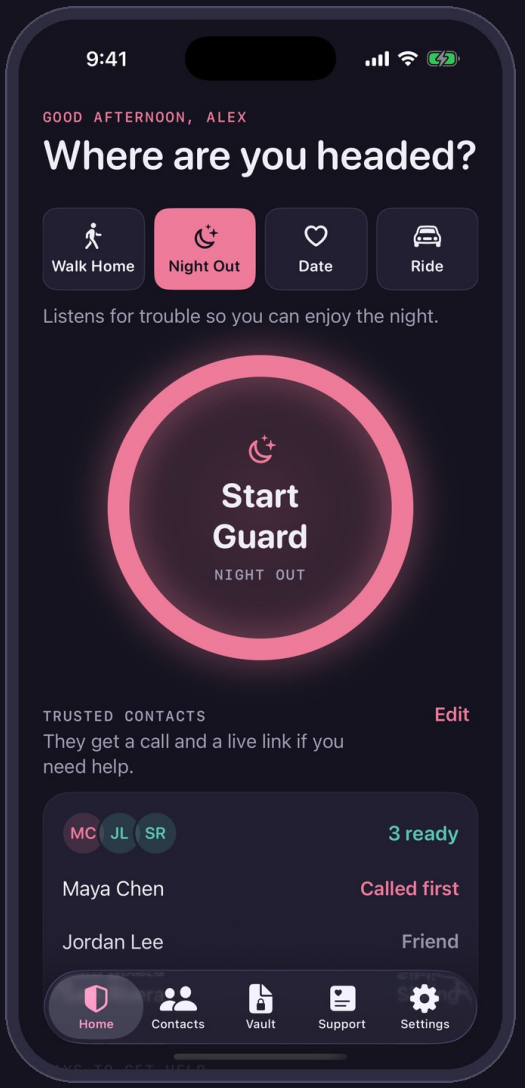
3

## Help comes even if you can't tap

Missed check-ins, distress sounds or a dead phone start the ladder.

SIGNALS, ON THE PHONE ONLY

- Screams, shouts, crying, glass breaking
- Your safe word or “get off me”
- A fall, then stillness
- Off route, or not where you said
- A missed “You okay?”
- Phone goes dark: the server notices

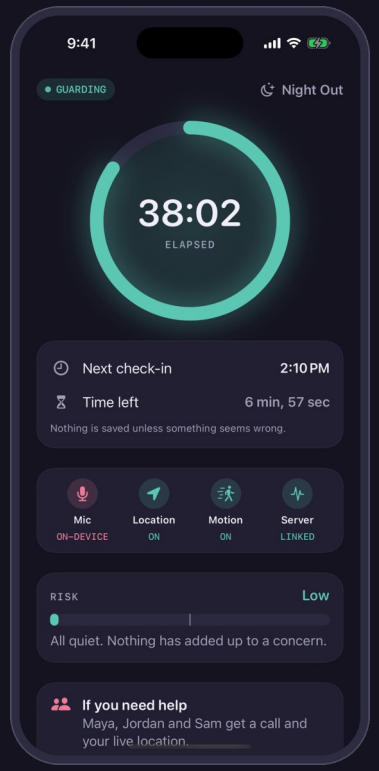



Five modes, one tap to start

HOW IT ESCALATES

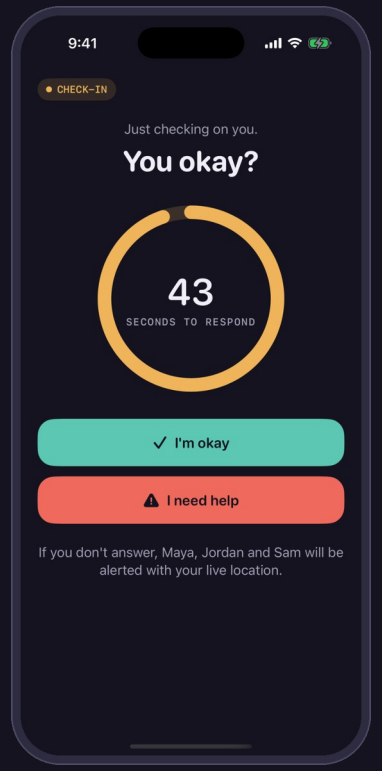
# One calm ladder, from “You okay?” to dispatch

## 0 · Guarding



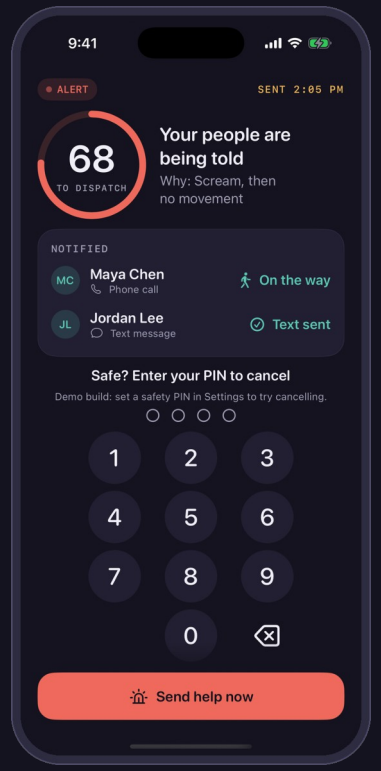

Audio stays in a 30 s memory buffer. Nothing is saved.

## 1 · Check-in



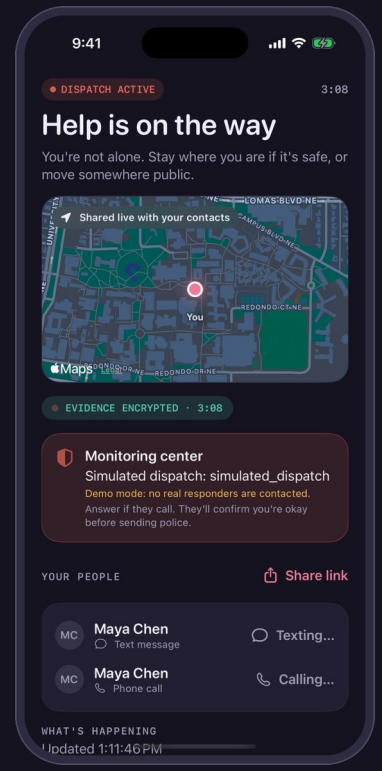

A quiet haptic: “You okay?” You have 45 seconds to tap.

## 2 · Alert




Your people get a call and a live map. Evidence is encrypted.

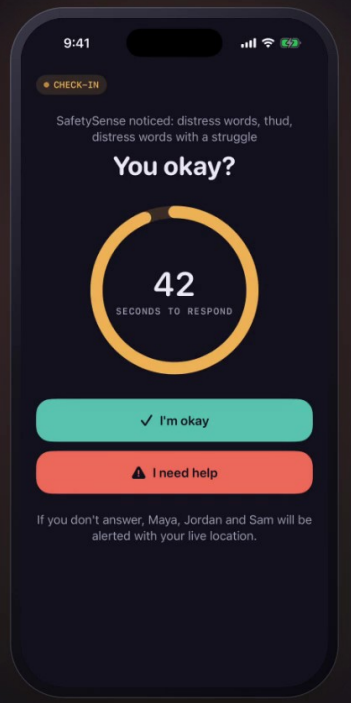
## 3 · Dispatch




After 90 s or “Send help,” a monitoring center calls, then sends police.

LIVE DEMO · RECORDED

# Night Out: a thud, then silence




REAL APP · IOS SIMULATOR · DEMO MODE · NO REAL RESPONDERS

0 · GUARDING
1 · CHECK-IN
2 · ALERT
3 · DISPATCH
SAFE

**“You okay?”**

45 seconds to answer. One tap if you're fine.

so it calls and texts your trusted people.

1. **Night Out guard on**
   Mic, motion and location are checked on the phone. Nothing saved.
2. **A thud, then “get off me”**
   Detected on the phone. “You okay?” appears.
3. **No answer in 45 s**
   Trusted contacts get a call, a text and a live link.
4. **Dispatch, then safe**
   Simulated monitoring step. The safety PIN cancels it.
5. **The contact's live link**
   Location, battery, and “I'm on my way.”

ORIGINALITY

# We built the basics. Then we connected the parts that work when you can’t.

| | UNM Safe | Noonlight | bSafe | Apple built-in | SafetySense |
|---|---|---|---|---|---|
| **Notices distress on its own** | – | Crash (paid) | – | Crash; fall on Watch | ✓ **Sound, motion, context** |
| **Acts when the phone goes dark** | – | – | – | Check In, timing varies | ✓ **Server heartbeat** |
| **Duress PIN (fake cancel)** | – | – | – | – | ✓ |
| **Date details sealed until needed** | – | Tinder notes | – | – | ✓ |
| **Professional dispatch** | Call UNMPD | ✓ | – | 911 call | ✓ **Noonlight** |
| **Voice trigger** | – | – | ✓ | Siri | ✓ **Offline safe word** |
| **Live location to a friend** | ✓ | ✓ | ✓ | ✓ Check In | ✓ |

*Others have pieces: Centinela detects but doesn’t dispatch; aegis+ dispatches but doesn’t detect. We found no product that documents the full chain.*

DESIGN AND EXPERIENCE

# Designed for the worst night of someone's life

### Calm, never blaming

“Whatever happened, it isn't your fault.” Plain words, no alarms in the copy.

### Practice before it's real

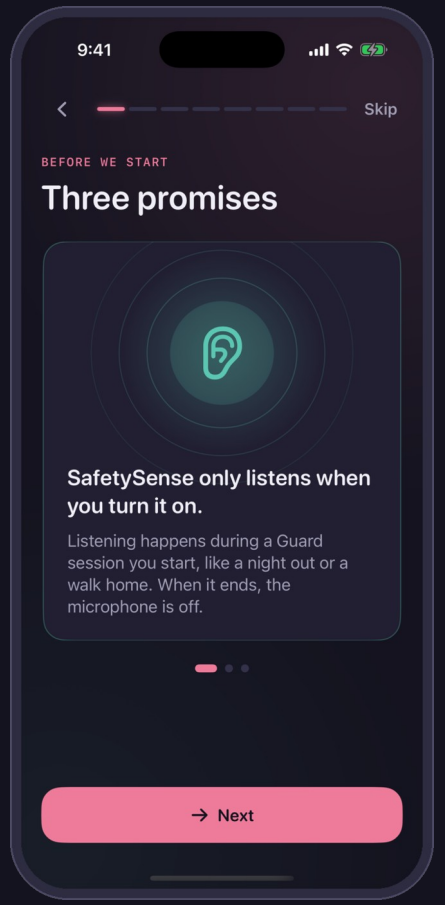
Onboarding ends with a drill. Contacts see it marked TEST.

### Safe if someone grabs the phone

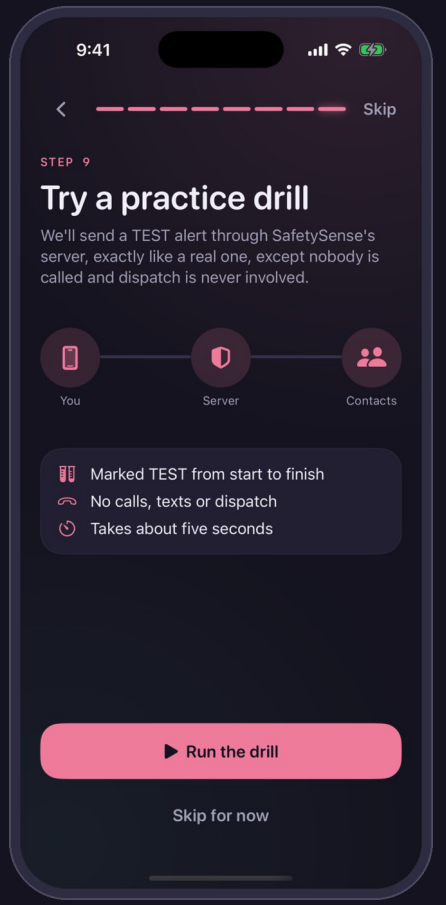
Duress PIN, Face ID vault, a discreet app icon.

### Nothing forced afterward

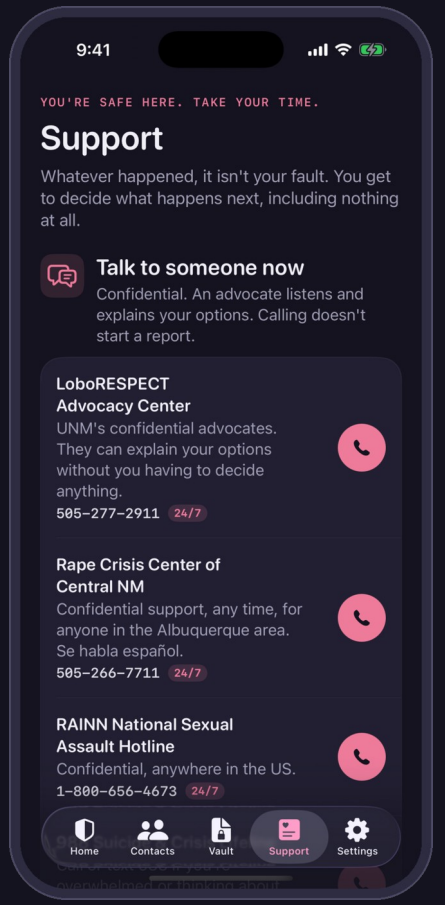
Advocate, medical care, reporting, or keeping it private. Your choice.




Three promises




Practice drill




Support, after

PRIVACY AND SECURITY

# Private by default. Evidence only when it matters.

### Guarding
30 s audio buffer in RAM only, overwritten constantly

### Alert fires
That buffer plus live audio is captured

### Encrypted on the phone
AES-GCM (CryptoKit), key in the Keychain

### Uploaded while recording
Server stores ciphertext only. Survives a smashed phone

### No ad or analytics SDKs
No third-party trackers at all.

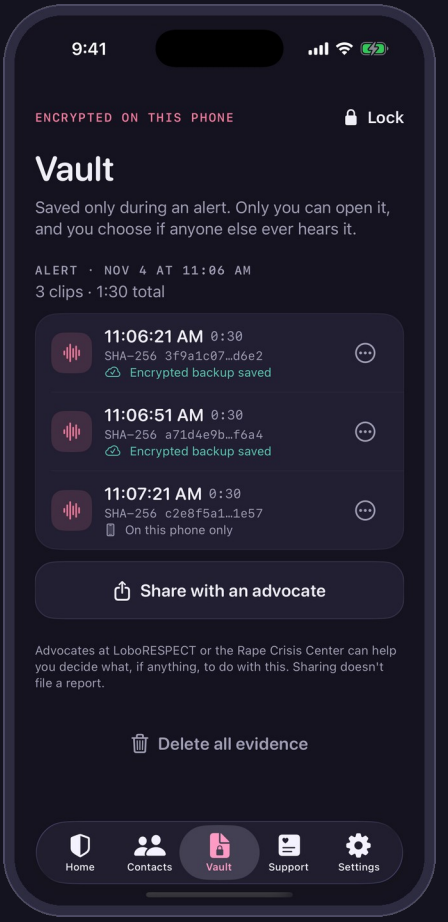
### Duress PIN
Looks like a cancel. Help still comes.

### Expiring live links
Contacts see your location only during an alert.

### Sealed date details
Shown to contacts only if an alert fires.

### Delete everything
Location is deleted after 30 days by default.

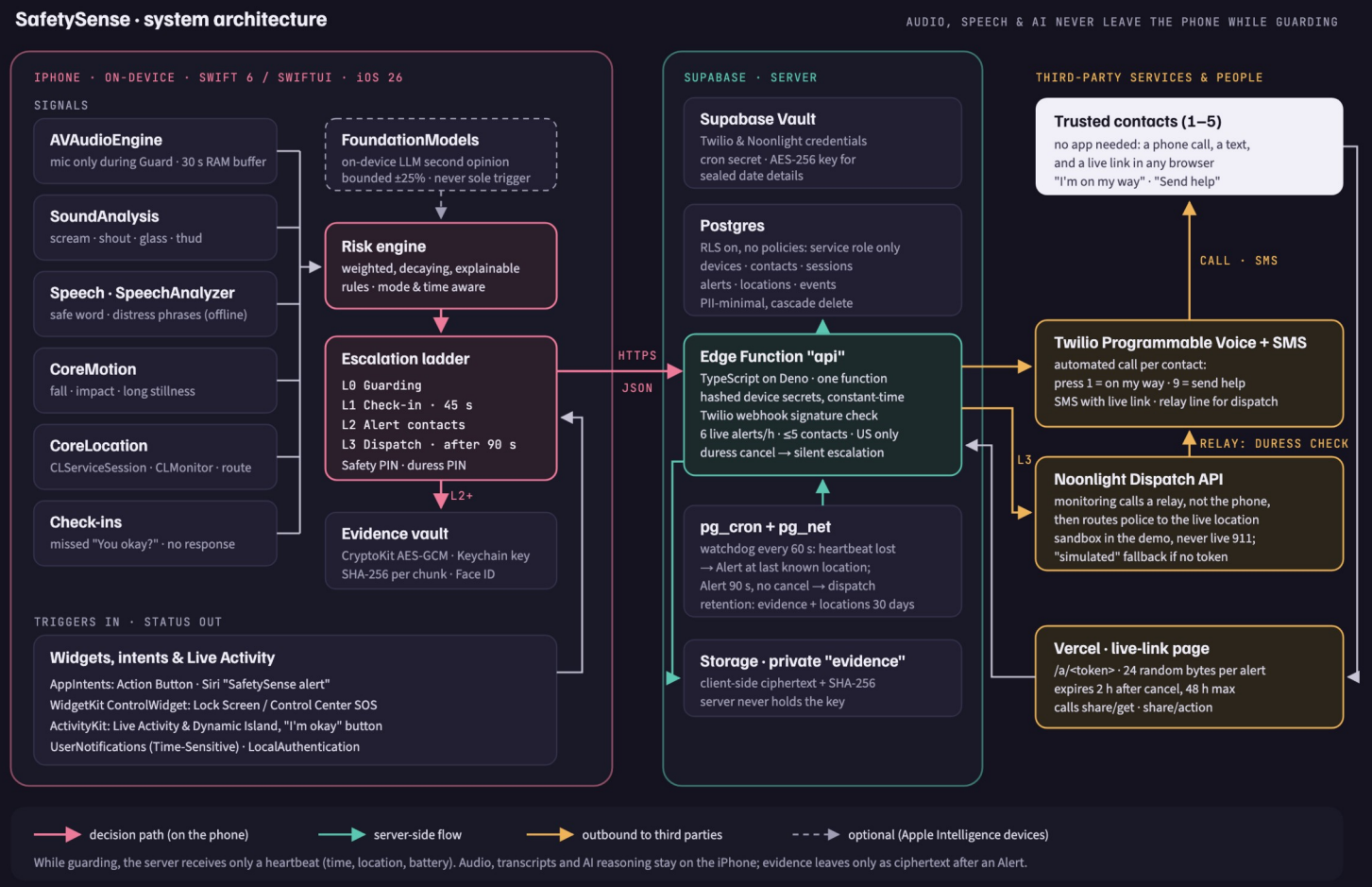
### SHA-256 per clip
Timestamped and hashed, so tampering shows.




Vault: Face ID, hashed clips

TECHNOLOGY

# Real app, real backend, AI that stays on the phone



PATHWAY TO SATISFACTION

# How we'll know it works

**< 1**

false alert per 10 hours of Night Out guarding

**< 60 s**

from a scream to contacts being called

**< 8%**

battery per hour while guarding

**≥ 80%**

of new users finish onboarding

ROLLOUT

**Now**

Working iOS prototype on a live backend

**Next**

Usability tests with UNM and CNM students; advocates review the copy

**Pilot**

TestFlight pilot with a student organization

**Partner**

Campus partnership: orientation, Women's Resource Center, UNMPD

SAFETYSENSE

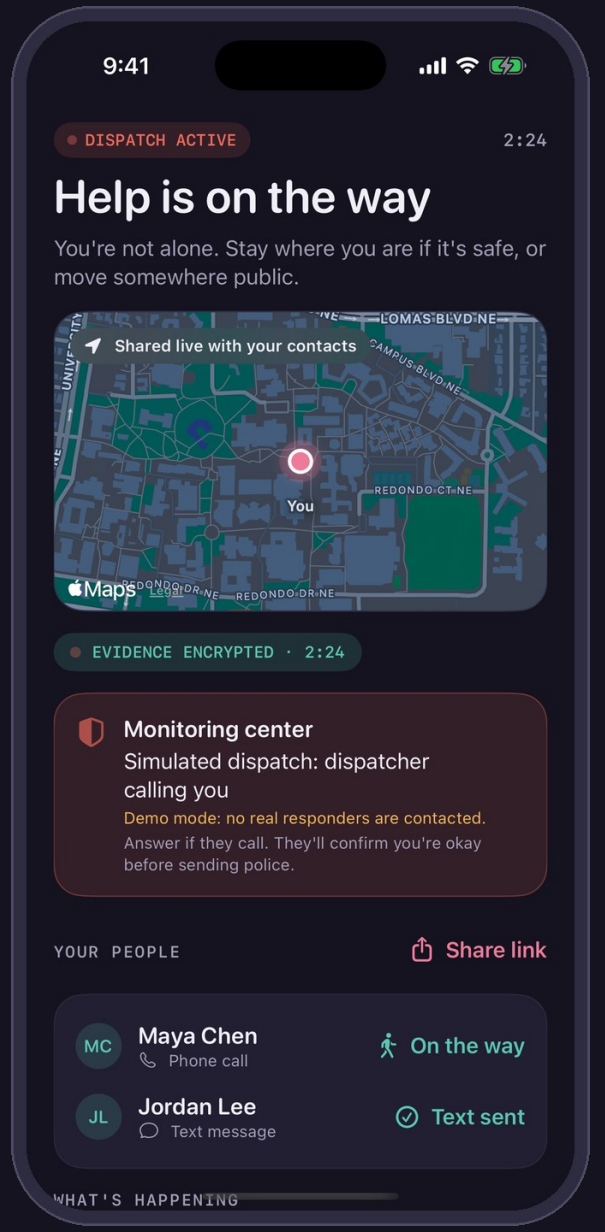
# When you can't call for help, SafetySense does.

Help us pilot it with students at UNM and CNM.

**George Teifel** UNM **Mary Teifel** CNM

Thank you. We'd love your questions.

*SafetySense supplements 911. It never replaces it.*



# Likely judge questions: trust and tech

## What about false alarms?

Escalation needs a combination of signals that fade over time, and the first step is always a quiet “You okay?” that one tap clears. Contacts come before dispatch, and the monitoring center's call is forwarded to you to confirm before sending police. Target: under 1 false alert per 10 hours.

## Won't listening kill the battery?

It only listens during a Guard session you start, not all day. Sensing runs on Apple's on-device frameworks, and the server heartbeat is one small request a minute. Our target is under 8% per hour, and we're measuring it on an iPhone 16e.

## Is it recording my conversations?

No. Audio sits in a 30-second memory buffer that overwrites itself and is never written to storage. Only an Alert keeps audio, and that's encrypted on the phone. Transcription and AI run on the device. The server sees ciphertext only.

## Will Apple approve it?

We follow the rules rather than fight them: listening only during user-started sessions with a visible Live Activity, no auto-dialing 911 (iOS doesn't allow it), and dispatch through Noonlight, an established partner. Permissions are explained when they're asked for.

# Likely judge questions: cost, law, business

## What does it cost to run?

A Twilio call or text costs a cent or two, and only during alerts. Supabase is free at pilot scale and about $25 a month after that. Dispatch is priced per alarm through Noonlight's partner program, which we'd confirm before launch.

## Why not Apple Check In or UNM Safe?

Both are good, and we link to UNM Safe. Check In notices when you don't arrive or a timer runs out. UNM Safe needs you to tap. Neither listens for distress or has a duress PIN. We cover the moments you can't reach your phone.

## Is recording legal?

Audio is kept only during an alert, and the user is a party to the conversation. New Mexico is generally read as a one-party-consent state (NMSA 30-12-1), but we'd get a legal review before launch. Clips belong to the user and are shared only if they choose.

## Business model and partners?

Free for students. Proposed revenue is a campus licence paid from safety and Title IX budgets, plus a paid plan for non-students. Partners we hope to work with: LoboRESPECT, the Women's Resource Center, UNMPD and CNM Security.

# Likely judge questions: edge cases

## What if nobody answers, or there's no signal?

If no contact responds in 90 seconds, dispatch takes over. If the phone loses signal or dies mid-session, the server notices the missing heartbeats and alerts contacts with the last known location.

## Could an abuser use it to track someone?

Location is shared only while you're guarding or in an alert, and only you can start that. Contacts can't pull your location on demand, and live links are built to expire after the alert.

## What about older iPhones?

The Action Button needs a newer iPhone, so there's also a Lock Screen control, Siri, a spoken safe word and a hold-to-release SOS. Without Apple Intelligence the app runs on rules only. It needs iOS 26.

## Android? Other campuses?

We started with iOS so we could use Apple's on-device AI. The backend and the contacts' live link already work from any phone or browser. Android and other campuses come after the student pilot.